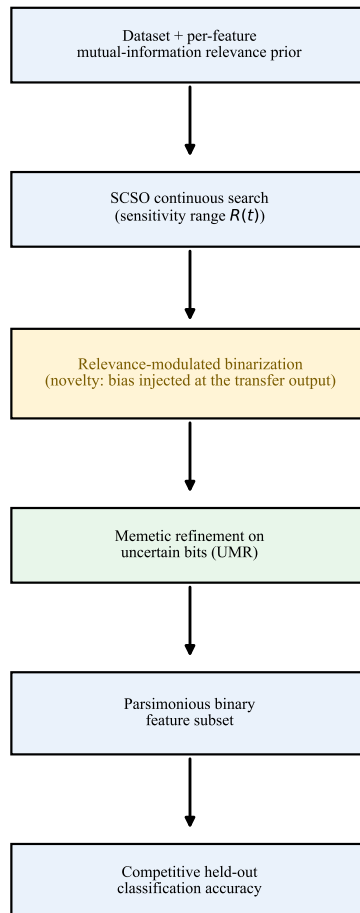


Graphical Abstract

Relevance-Guided Binarization for Parsimonious Feature Selection with Sand Cat Swarm Optimization

Bui Quang Huy, Duong Minh Son

**RG-SCSO: relevance-guided binarization for
parsimonious feature selection**



Highlights

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- Transfer-function binarization discards swarm search detail; we name it washout
- RG-SCSO injects per-feature MI relevance into the binarization decision of SCSO
- Best mean accuracy and second-smallest subsets across 18 datasets at equal budget
- Ablation localizes the gain to the binarization step, not init or the objective
- Leak-free gain holds across KNN/SVM/RF wrappers; bounded on p»n gene-expression

Relevance-Guided Binarization for Parsimonious Feature Selection with Sand Cat Swarm Optimization

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Abstract

Wrapper feature selection with swarm intelligence typically searches in continuous space and crosses into the binary domain through a fixed transfer function – a feature-agnostic quantization that discards the continuous operators’ fine adjustments, a failure mode we identify and name washout. We diagnose washout as the reason continuous-space enhancements stop helping at the binarization boundary, and address it at that boundary: RG-SCSO replaces the fixed transfer with a per-feature, relevance-modulated binarization in which a mutual-information field biases each feature’s bit-flip probability, so informative features resist removal and noise resists inclusion, while SCSO’s continuous search is left otherwise unchanged. The design targets low- to medium-dimensional wrapper feature selection, where the practical goal is a smaller subset at matched accuracy. Evaluated on 18 datasets under a fixed-budget, leak-free protocol that computes the relevance prior, the search, and the cross-validated fitness only on the training partition, RG-SCSO attains the best mean held-out accuracy of the seven metaheuristics compared while selecting the second-smallest feature subsets of any method, trailing only the less-accurate COA. A controlled ablation localizes the gain to the binarization decision – not initialization or the objective – and it persists across classifier wrappers and relevance priors, with more stable subsets than the base search. On the two gene-expression ($p \gg n$) sets a classical LASSO baseline is preferable, bounding the method’s envelope to data without that extreme structure.

Keywords: Feature selection, Sand Cat Swarm Optimization, binary optimization, relevance-guided binarization, parsimony

1. Introduction

Feature selection removes irrelevant and redundant features to improve classifier accuracy, reduce overfitting, and lower computational cost, a payoff that is greatest for high-dimensional, small-sample problems such as gene-expression classification, where the number of features exceeds the number of samples by orders of magnitude [1, 2]. Wrapper selection, which scores a subset by the performance of a downstream classifier, is frequently cast as a combinatorial problem solved by swarm-intelligence metaheuristics, including grey wolf [3], particle swarm [4], whale [5], and several recent optimizers [6, 7, 8]. Most such methods were conceived for continuous optimization and are adapted to the binary space through a transfer function, most commonly an S-shaped or V-shaped map, that converts a real-valued position into a selection probability [9]. We argue that this retrofit contains a structural weakness: the transfer function is fixed and feature-agnostic, applied identically to every dimension, so the incremental adjustments a continuous-space operator makes are collapsed by the squash-and-threshold step before they can influence the retained subset. We refer to this loss as washout, and show in Methods below that it is not a thought experiment: four well-motivated continuous-space enhancements of SCSO [10] failed to beat the base algorithm under an identical feature-selection protocol (0 wins, 1 loss, 17 ties by Wilcoxon signed-rank; Supplementary Table S5, preliminary washout study).

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Existing studies have explored relevance-guided initialization, objective weighting, and adaptive binary transfer mechanisms; however, the explicit injection of a per-feature relevance prior into the binarization operator of SCSO-based feature selection remains insufficiently investigated (Section 2 positions this gap against each cited SCSO-family work in detail).

We close this gap with RG-SCSO: a per-feature, relevance-modulated binarization in which a mutual-information relevance field biases each feature’s bit-flip probability, turning a knowledge-agnostic quantization step into a knowledge-carrying operator. SCSO’s continuous search, including its sensitivity range, is retained unchanged; the novelty resides entirely in the binarization (Fig. 1). The method is designed for the regime where wrapper feature selection is most often applied in practice – low- to medium-dimensional problems in which the goal is a smaller, more deployable feature subset at matched accuracy – and we evaluate it, and bound it, against that intended use rather than claiming universal dominance; on extreme $p \gg n$ gene-expression data a classical embedded selector remains preferable, a boundary we report as a scope of the design, not a hidden caveat. This paper makes four contributions. First, we identify washout as a mechanistic failure mode and derive a diagnostic bound, together with a cumulative extension linking it to discrete transition dynamics. Second, we propose RG-SCSO, whose ablation-confirmed centerpiece is relevance-modulated sensitivity (RMS), supplemented by a smaller, budget-neutral memetic refinement step (UMR) that a sensitivity sweep shows contributes far less than RMS; an online-learning variant of the relevance field is examined and pruned entirely by ablation. Third, we evaluate under a pre-specified, fixed-budget, leak-free protocol that denies the relevance prior any access to test labels. Fourth, we report a full statistical treatment, a component ablation, and a size-fair enrichment analysis correlating the observed parsimony with relevance guidance.

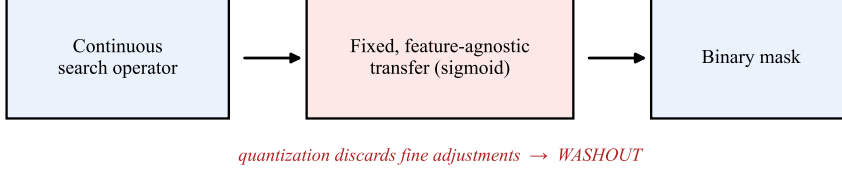
1.1. Research questions

This study is organized around four research questions, each answered directly by a specific part of Results or Discussion below.

- RQ1. Does relevance-modulated binarization improve held-out feature-selection performance under a fixed computational budget? Addressed in Section 3.1 (Table 2, Fig. 7).
- RQ2. Can RG-SCSO reduce the number of selected features while preserving competitive generalization accuracy? Addressed in Section 3.2.
- RQ3. Is direct relevance injection at the binarization interface more effective than relevance-guided initialization or objective weighting? Addressed in Section 3.4 (Table 4).
- RQ4. How robust is the proposed mechanism across classifiers, relevance priors, and dataset dimensionalities? Addressed in Section 3.6 (Table 7) and Discussion.

The remainder of the paper reviews existing approaches along three axes: (i) binary transfer functions in swarm-based feature selection, (ii) relevance-guided search mechanisms, and (iii) SCSO variants (Section 2).

(a) Conventional continuous-to-binary pipeline



(b) RG-SCSO: sensitivity range → per-feature bit-flip

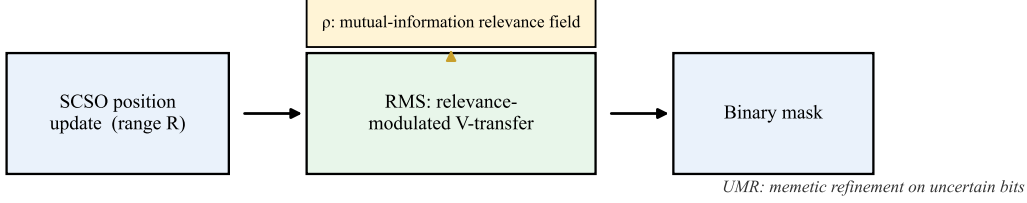


Figure 1: Conceptual overview. (a) The conventional pipeline, where continuous-operator adjustments are collapsed by a fixed, feature-agnostic transfer (washout). (b) RG-SCSO, where a per-feature, relevance-modulated binarization replaces the feature-agnostic transfer, biasing each feature’s bit-flip probability by a mutual-information relevance field, followed by memetic refinement on uncertain bits.

2. Related work

2.1. Binary transfer functions in swarm-based feature selection

Swarm-intelligence wrapper selectors, including grey wolf [3], particle swarm [4], whale [5], and several recent optimizers [6, 7, 8], are conceived for continuous optimization and cross into the binary search space through a transfer function, most commonly an S-shaped or V-shaped map [9]. A substantial recent line of work makes this interface adaptive rather than static: time-varying transfer functions that reshape the squashing curve as the search progresses [11, 12], and fitness- or history-driven schemes that tune it from the swarm’s own progress [13, 14]. Crucially, all of these remain feature-agnostic: the same (possibly time-varying) transfer is applied identically to every dimension regardless of that feature’s relevance, the structural weakness this paper terms washout (Introduction, Methods). RG-SCSO is orthogonal to this line – it modulates the binarization per feature by a relevance prior, and could in principle be combined with an adaptive transfer rather than replacing one.

2.2. Relevance-guided mechanisms in feature selection

Filter criteria such as mutual information and mRMR [1, 2] encode problem knowledge cheaply but are decoupled from the wrapper search; memetic hybridization [15] adds local refinement without addressing the binarization interface. A large family of knowledge-guided metaheuristics injects filter information upstream of the binary decision, into initialization or the objective. In other swarm families this takes the form of correlation- or mutual-information-guided population initialization [16, 17], filter-guided PSO for cancer-genome selection [18], and memetic hybridization of binary Harris-hawks and whale operators [19]; a parallel, deep-learning-oriented direction drives selection from post-hoc attribution or attention, for example SHAP-guided deep feature selection [20]. All of these bias where the search starts, what it optimizes, or which features a separately trained model highlights, yet leave the binarization operator itself knowledge-agnostic. RG-SCSO differs by injecting the relevance signal at the binarization interface itself rather than upstream of it; the signal-position experiment in Results (Table 4) tests this design choice directly against both the initialization and objective alternatives.

Table 1: Positioning of RG-SCSO against recent SCSO-family literature cited in this paper. The four 2024–2025 continuous-search variants (IMSCSO, SCSO+Lens-OBL+SSA, Improved SCSO, MESCSO) improve exploration/exploitation dynamics in continuous space but are global optimization studies, not feature selectors, so they are not directly comparable under this paper’s binary FS protocol; they are cited to establish that none of them touches the binarization interface itself. The three binary SCSO feature selectors (bSCSO and two further variants) are directly comparable and are included as same-family baselines (Discussion; Supplementary Information).

Method	Year	What it modifies	Comparable to this protocol?
Seyyedabbasi & Kiani [10]	2022	Base continuous SCSO (no FS)	Baseline (Table 1)
bSCSO [21]	2023	Binary wrapper FS, standard transfer	Same-family baseline (Discussion)
Binary SCSO (biomedical) [22]	2023	Binary wrapper FS, standard transfer	Same-family baseline (Discussion)
Adaptive SCSO [23]	2024	Binary wrapper FS, standard transfer	Same-family baseline (Discussion)
IMSCSO [24]	2024	Continuous search (multi-strategy)	No – global optimization only, no FS
SCSO+Lens-OBL+SSA [26]	2024	Continuous search (lens-OBL init)	No – global optimization only, no FS
Improved SCSO [27]	2024	Continuous search dynamics	No – global optimization only, no FS
MESCSO [25]	2025	Continuous search (multi-strategy)	No – global optimization only, no FS

Positioning against relevance-injecting binary GWO/WOA/HHO variants. The closest competitors are binary grey-wolf, whale, and Harris-hawks selectors that already carry a relevance signal: MOBGWO-GMS initializes its population from feature correlation and guides mutation with it [16], the enhanced Harris-hawks optimizer seeds the search with filter-ranked features [17], and hybrid binary whale/Harris-hawks memetics combine two operators for stronger local search [19]. Each injects relevance at initialization, mutation, or hybridization; none makes the per-feature binarization operator itself relevance-aware, which is the specific interface RG-SCSO modifies and the reason its parsimony gain (Results) is not reproduced by these variants’ upstream injections.

2.3. Sand Cat Swarm Optimization and its variants

SCSO [10] is a comparatively recent continuous metaheuristic. The existing binary SCSO feature selectors [21, 22, 23] apply standard transfer functions without a relevance-aware binarization step, and the broader recent SCSO literature that adds chaotic initialization, differential mutation, or hybridized search strategies [24, 25, 26, 27] improves continuous-space search dynamics while leaving the binarization interface itself untouched – across the SCSO-family literature we screened, we found none that makes the binarization operator itself per-feature and relevance-aware. A per-reference audit of the binarization mechanism (transfer function; whether relevance enters at the binarization step or only upstream) is released with the code; notably, the closest binary SCSO selector that uses relevance injects it at initialization (a differential-expression pre-filter), not at the binarization operator, which is precisely the distinction our signal-position experiment (Section 3.4) isolates. Table 1 below positions each of these works against this claim directly.

2.4. Literature-positioning summary

Table 1 makes this gap concrete across every SCSO-family work cited above: the three directly comparable binary feature selectors apply a standard, feature-agnostic transfer with no relevance-aware step, and the four continuous-search variants improve exploration/exploitation dynamics without touching the binarization interface at all. None of the eight prior works positioned here makes the binarization operator itself per-feature and relevance-aware – the interface RG-SCSO modifies.

Table 2: Held-out generalization: mean accuracy, with mean number of selected features in parentheses, on the outer 20% hold-out over 30 runs (relevance prior, search, and CV fitness fit on the 80% training split only; dataset feature counts are given in Supplementary Table S1). Standard deviations are omitted here for compactness; per-run raw results, from which they can be recomputed exactly, are in the repository (Data availability). Bold = best accuracy per dataset.

Dataset	RG-SCSO	AOA	COA	SCSO	RIME	PSO	GWO
BreastEW	0.9630 (5.2)	0.9679 (8.7)	0.9487 (4.7)	0.9509 (4.4)	0.9521 (3.9)	0.9428 (4.8)	0.9511 (4.1)
ColonCancer	0.7615 (555.1)	0.7410 (1958.2)	0.7103 (534.6)	0.7410 (1002.3)	0.7308 (999.3)	0.7333 (1006.6)	0.7513 (1007.0)
Diabetes	0.7229 (4.6)	0.7279 (7.4)	0.7000 (4.7)	0.7030 (4.1)	0.6883 (3.4)	0.6695 (3.9)	0.6708 (3.9)
GermanCredit	0.7260 (8.4)	0.7353 (19.9)	0.7020 (11.2)	0.7038 (10.0)	0.7048 (9.8)	0.6980 (10.5)	0.6948 (9.7)
HeartDisease	0.8089 (5.9)	0.8261 (12.8)	0.7817 (7.1)	0.7661 (6.6)	0.7633 (6.3)	0.7367 (6.6)	0.7567 (6.3)
IonosphereEW	0.8864 (8.0)	0.8418 (33.9)	0.8540 (11.8)	0.8549 (16.5)	0.8531 (15.8)	0.8418 (16.6)	0.8385 (16.5)
KrVsKpEW	0.9518 (12.2)	0.9289 (35.0)	0.8383 (17.4)	0.8351 (17.5)	0.7808 (16.4)	0.7908 (17.1)	0.7120 (17.2)
Leukemia	0.9733 (957.4)	0.9622 (3549.2)	0.9511 (619.4)	0.9533 (1770.7)	0.9644 (1778.9)	0.9600 (1780.8)	0.9578 (1788.6)
Lymphography	0.7978 (7.0)	0.8167 (17.9)	0.7611 (10.1)	0.7711 (9.1)	0.7556 (8.7)	0.7611 (8.6)	0.7356 (8.0)
M-of-n	0.9960 (6.3)	0.8712 (12.8)	0.7882 (7.2)	0.7733 (7.2)	0.7480 (6.6)	0.7263 (6.1)	0.6985 (6.2)
Parkinsons	0.9051 (6.6)	0.9026 (21.7)	0.8872 (10.3)	0.8812 (11.2)	0.8812 (10.8)	0.8718 (11.0)	0.8761 (10.8)
Sonar	0.8079 (16.7)	0.8024 (59.9)	0.8008 (29.0)	0.7976 (29.6)	0.7897 (29.7)	0.7944 (30.6)	0.8024 (29.8)
SpectEW	0.8204 (7.8)	0.8099 (21.4)	0.8031 (9.9)	0.8012 (11.0)	0.7951 (10.0)	0.8074 (11.0)	0.7932 (10.8)
TicTacToe	0.8212 (7.7)	0.7856 (8.4)	0.6924 (5.9)	0.6462 (4.5)	0.6538 (4.8)	0.6552 (4.6)	0.6241 (4.1)
Vote	0.9560 (3.0)	0.9220 (15.9)	0.9014 (5.7)	0.8837 (6.5)	0.9028 (7.9)	0.8915 (7.2)	0.9021 (8.3)
WDBC	0.9667 (12.5)	0.9649 (29.7)	0.9585 (14.1)	0.9561 (14.9)	0.9558 (15.0)	0.9532 (14.5)	0.9544 (15.0)
WaveformEW	0.8153 (15.8)	0.7571 (38.3)	0.7456 (21.0)	0.7352 (20.6)	0.7170 (20.2)	0.7238 (20.8)	0.6863 (19.0)
Zoo	0.9079 (7.8)	0.9190 (15.7)	0.8460 (8.6)	0.8540 (8.3)	0.8460 (8.0)	0.8476 (8.3)	0.8270 (7.8)

3. Results

3.1. Held-out generalization

RQ1 asks whether relevance-modulated binarization improves held-out performance under a fixed computational budget; this subsection answers it directly. For each dataset, algorithm, and independent run we draw an outer stratified 80/20 split. The relevance prior, the search, and the cross-validated fitness are computed exclusively on the 80% training partition; the selected subset is evaluated once on the untouched 20% hold-out, on which a fresh k -NN classifier (standardized on the training partition) reports accuracy, so the relevance prior never has transductive access to the test labels. Table 2 reports held-out accuracy over all seven algorithms and 18 datasets, at a fixed evaluation budget of $\text{max_nfe} = 15000$ for every algorithm compared. RG-SCSO attains the best average Friedman rank (1.33, ahead of the second-placed AOA at 2.03; $\chi^2=67.25$, $p < 10^{-3}$). A Holm-corrected Wilcoxon signed-rank test across all pairwise comparisons gives RG-SCSO 64 significant wins, 1 loss, and 43 ties; the only close competitor is AOA, over which RG-SCSO holds the higher mean accuracy and a consistent positive aggregate difference (median $|d| = 0.30$). The more conservative Nemenyi post-hoc, however, does not separate the two (Fig. 7), so we read this as a modest aggregate edge rather than a statistically conclusive superiority over AOA. Over the 18 datasets the paired accuracy difference against AOA is $+0.017$ (95% bootstrap CI $[+0.003, +0.034]$, excluding zero), confirming a small but consistent aggregate edge; against every other baseline the paired difference is larger (all 95% CIs excluding zero). Fig. 7 visualizes this held-out ranking as a critical-difference diagram: the Nemenyi test, a more conservative simultaneous comparison than the pairwise Wilcoxon test above, does not separate RG-SCSO from AOA, but places both above a single indistinguishable cluster comprising the remaining five algorithms.

3.2. Feature-subset parsimony

RQ2 asks whether RG-SCSO can reduce feature count while preserving competitive accuracy; this subsection answers it directly. Parsimony, more than any accuracy margin, is RG-SCSO’s defining property. Table 2 shows it attains the best mean accuracy of any method tested (0.866) while selecting the second-fewest features on the held-out setting, on average 91.6 against 164.2 for base SCSO and 325.9 for AOA. Only COA selects fewer (74.0 on average), and it does so at 0.051 lower mean accuracy, so RG-SCSO is the most compact method that does not trade away accuracy to get there. On ColonCancer it retains 563 of 2000 features versus 1991 for AOA, at higher accuracy. This advantage survives every stress test

we run: against optimizers carrying published adaptive transfers RG-SCSO does not lead on accuracy yet still selects 26–43% fewer features, and against same-family binary SCSO selectors [21, 22, 23] it is 42% smaller at comparable accuracy. The dimensionality reductions are too small relative to these datasets’ modest sample counts (50–455 training instances) for a wall-clock inference saving to be resolvable above call overhead; a projected, reconstructed inference-cost analysis at a synthetic deployment-scale workload, a controlled complexity demonstration rather than a measurement on the benchmark’s own test sets, is reported in Supplementary Information instead of here.

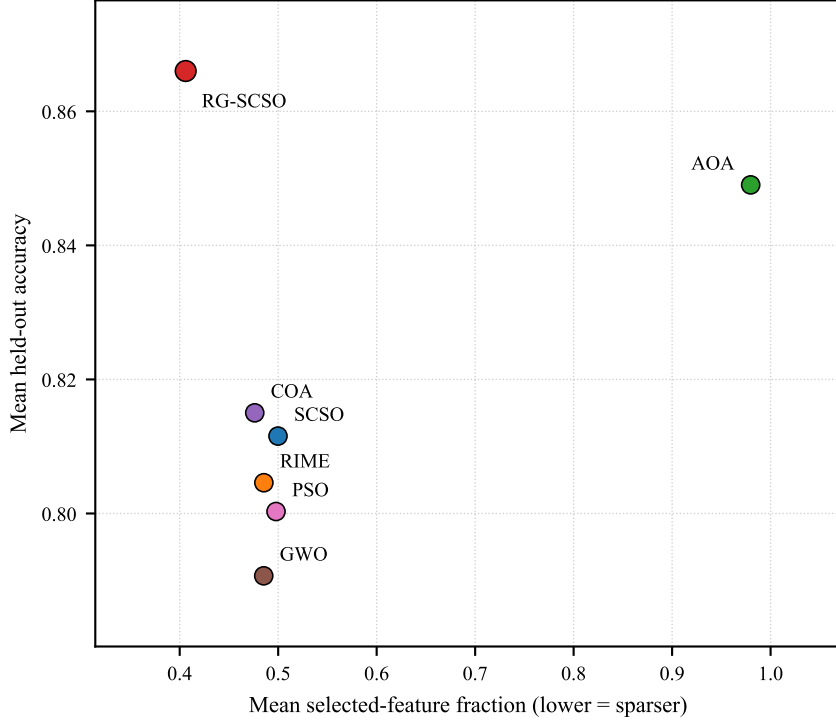


Figure 2: Mean held-out accuracy vs. mean selected-feature fraction, averaged across all 18 datasets, one point per algorithm. RG-SCSO occupies the top-left corner, jointly the highest accuracy and the sparsest subsets of all seven algorithms tested – it is not merely competitive on one axis at the cost of the other. AOA attains the second-highest accuracy but selects on average 98% of available features, essentially no feature selection; COA is the second-sparsest method but at a distinctly lower accuracy than RG-SCSO. In numbers, RG-SCSO’s mean operating point is 0.866 accuracy at 41% of features; AOA reaches 0.849 at 98% of features, and COA 0.815; mean accuracy across the seven algorithms spans 0.791–0.866.

3.3. Ranking and statistical significance

Table 3 gives the in-sample average rank (1.00 for RG-SCSO) and the Holm-significant win/tie/loss against each baseline (the corresponding critical-difference diagram is Supplementary Fig. S6; Fig. 7 in the main text gives the held-out counterpart). These are paired per-dataset comparisons, not independent trials: across the 108 dataset-baseline comparisons RG-SCSO records 107 significant wins, 1 non-significant tie, and 0 significant losses after Holm correction, with predominantly large effect sizes (median $|d| = 2.15$; 96% exceed 0.8). These in-sample effect sizes are an optimistic upper bound: they contract to the small-to-moderate held-out range above once the relevance prior is denied access to test labels, whereas the ranking itself is preserved.

3.4. Ablation and mechanism

We started from a three-component design and tested each part by removal (Table 4), judging significance with a paired Wilcoxon signed-rank test (Holm-corrected across datasets). RMS is the strongest: removing

Table 3: Average accuracy rank across the 18 datasets, the Holm-significant win/tie/loss of RG-SCSO against each baseline (Wilcoxon signed-rank, $\alpha = 0.05$), and the median |Cohen’s d | effect size over the significant-win comparisons against each baseline.

Algorithm (year)	Avg. rank	W/T/L vs. RG-SCSO (Holm)	Median $ d $
RG-SCSO (ours)	1.00	–	–
AOA (2021)	2.89	17/1/0	3.42
SCSO (2022)	3.94	18/0/0	1.49
COA (2023)	4.11	18/0/0	1.87
RIME (2023)	4.44	18/0/0	2.08
PSO (1995)	4.94	18/0/0	2.32
GWO (2014)	6.67	18/0/0	2.33

Table 4: Component ablation: mean accuracy over 30 runs on five datasets. [†] significantly worse than Full (paired Wilcoxon signed-rank, Holm-corrected $p < 0.05$). “– ORL” is the final RG-SCSO (RMS+UMR); ORL removal never degrades accuracy, so ORL is not retained.

Configuration	ColonCancer	Leukemia	Sonar	WDBC	Zoo
Full (RMS+ORL+UMR)	0.8831	0.9860	0.8996	0.9793	0.9801
– RMS	0.8409 [†]	0.9856	0.8970	0.9795	0.9804
– ORL (final)	0.8809	0.9860	0.8940	0.9794	0.9808
– UMR	0.8815	0.9860	0.8880 [†]	0.9789	0.9804
– all three	0.8441	0.9860	0.8857	0.9781	0.9808

it costs 4.23 accuracy points on ColonCancer ($d = 3.21$, Holm $p < 0.001$). UMR is also load-bearing (1.16 points on Sonar), whereas ORL is not (degrades accuracy on only 0/5 datasets) and is therefore dropped; the final RG-SCSO comprises RMS and UMR only.

RQ3 asks whether direct relevance injection at the binarization interface is more effective than relevance-guided initialization or objective weighting. A dedicated signal-position experiment answers it: injecting the same mutual-information field elsewhere, at initialization or as an objective penalty, rather than at the binarization interface (both alternatives are reported in full in Supplementary Information). MI-weighted objective injection is significantly worse than the deployed RMS+UMR configuration on four of five datasets and tied on the fifth (Leukemia), supporting the original claim that the binarization interface is a more effective injection point than the objective function. MI-guided initialization is a harder comparison: significantly worse on ColonCancer and Sonar, but statistically indistinguishable on Leukemia, WDBC, and Zoo, and on Leukemia specifically it matches RG-SCSO’s accuracy while selecting roughly four times fewer features (245 vs. 940). The binarization interface is thus the most reliable injection point across datasets, not a uniformly superior one; on the most extreme $p \gg n$ dataset tested, a simpler injection at initialization meets or beats it on both accuracy and parsimony. Figure 3 traces this ladder: the mean accuracy is essentially flat for relevance at initialization (V1) or in the objective (V2) and rises only once the relevance field enters the binarization decision itself (V3), while parsimony is already gained upstream.

We also test whether relevance guidance makes RG-SCSO preferentially retain high mutual-information features. Because a subset of size $|S|$ overlaps the top- $|S|$ mutual-information features at a chance rate of $|S|/N$, we report a size-fair enrichment (Fig. 8), the fraction of selected features in the top- $|S|$ set divided by this chance level. RG-SCSO’s subset is enriched above chance on both gene-expression sets, whereas the relevance-agnostic SCSO sits at chance, evidence consistent with the relevance field driving the smaller and more accurate subsets, though enrichment alone is correlational. A direct causal test, detailed in Supplementary Information, permutes the field’s feature identities while preserving its value distribution: the permuted field yields no significant accuracy difference from the real field on three of five datasets, a significant but negligible-effect difference on Leukemia, and a clear difference only on ColonCancer, while inverting the field’s sign is significantly worse than both on every dataset. The relevance field’s direction and scale therefore matter consistently; the exact per-feature ranking within it matters demonstrably on only one of five datasets, a materially weaker causal claim than the enrichment analysis alone would suggest.

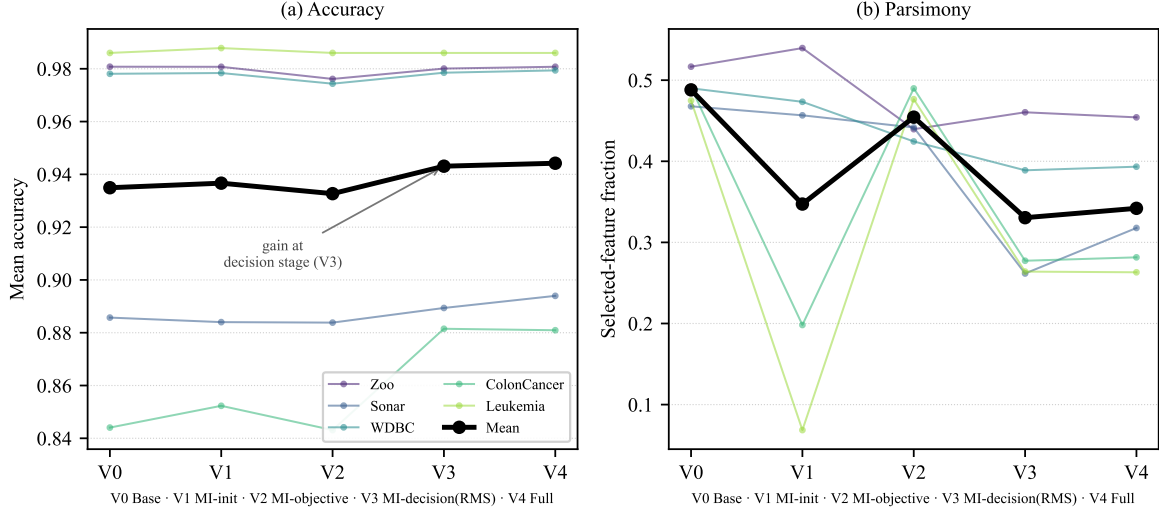


Figure 3: Controlled injection-point ablation of the relevance signal, from V0 (base SCSO, relevance absent) through V1 (relevance at initialization), V2 (relevance in the objective), V3 (relevance at the binarization decision, RMS) to V4 (full RG-SCSO, RMS+UMR), on five datasets spanning $p \gg n$ gene-expression to low-dimensional data, 30 runs per cell. (a) Mean accuracy: thin lines are per-dataset means, the bold line the across-dataset mean. The mean accuracy is essentially flat for V1 and V2 and rises only at V3, locating the accuracy effect at the binarization interface rather than upstream; the V4 UMR step adds little further accuracy (its benefit is in stability, Fig. 9). (b) Selected-feature fraction: parsimony is already gained at V1/V3, so the two effects arise at different injection points. The pattern is not uniform – on the most extreme $p \gg n$ set, initialization (V1) is competitive on both axes.

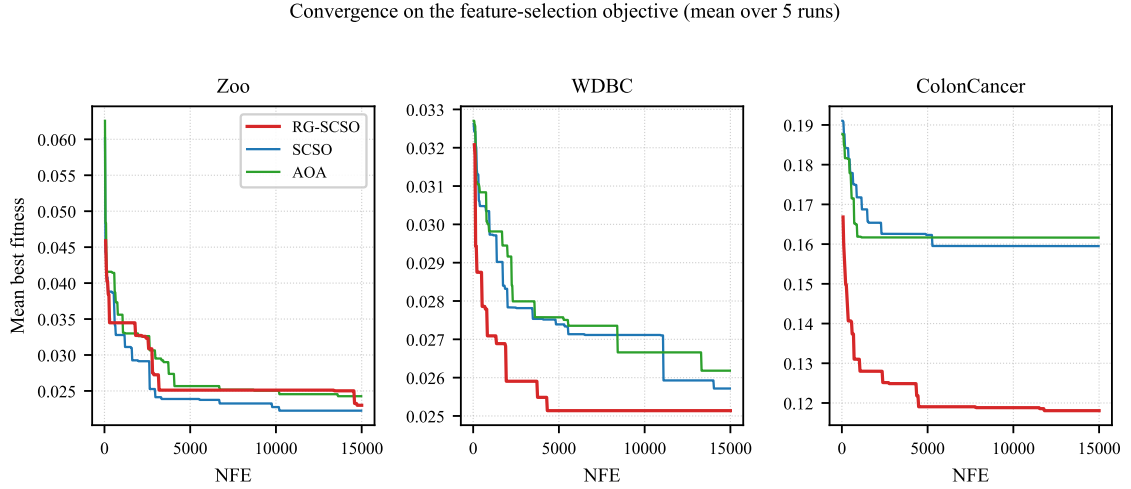


Figure 4: Mean best fitness versus iteration, RG-SCSO vs. SCSO vs. AOA, on Zoo (16 features), WDBC (30 features), and ColonCancer (2000 features), mean over 5 runs, on the actual feature-selection objective (illustrative, not a new statistical claim). On Zoo the three algorithms converge along essentially the same trajectory. On WDBC and ColonCancer, RG-SCSO both converges faster and plateaus at a lower (better) fitness than SCSO or AOA, which themselves plateau early at a distinctly worse value rather than continuing to close the gap with more iterations; the difference grows with dimensionality.

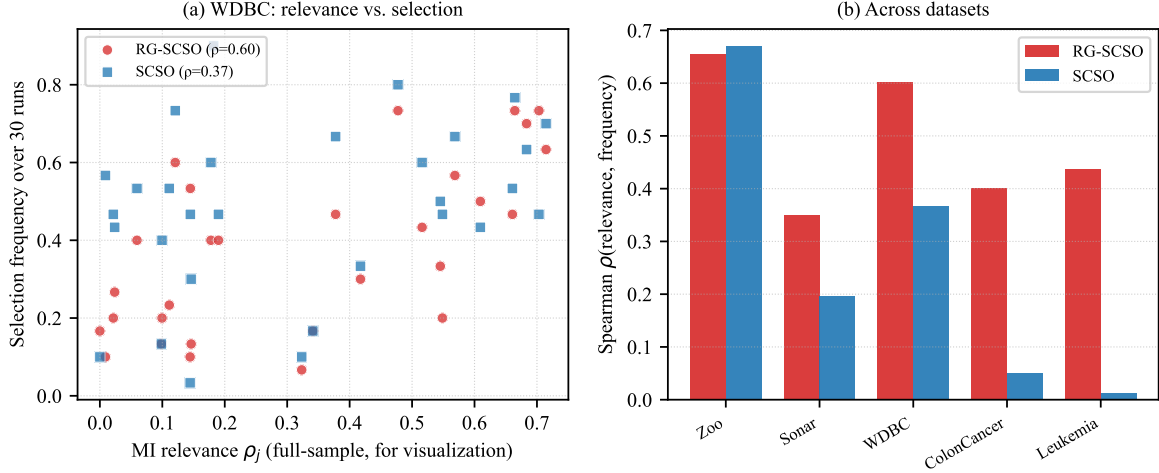


Figure 5: How relevance guidance translates into selection behaviour. (a) For WDBC, per-feature mutual-information relevance ρ_j (computed on the full sample for visualization only; the optimizer uses a train-only prior) versus the fraction of 30 runs in which each feature is selected. RG-SCSO’s selection frequency tracks relevance more closely than base SCSO, which does not use the prior (Spearman $\rho = 0.60$ vs. 0.37). (b) The same rank correlation across all five datasets: RG-SCSO is at least as relevance-aligned as SCSO on every set and markedly more so on the higher-dimensional ones (ColonCancer, Leukemia, WDBC), and essentially tied on low-dimensional Zoo where almost all features are informative. This association is correlational – RG-SCSO uses the prior by construction while SCSO does not – so the causal weight rests on the shuffled-prior control above, not on this alignment alone. The full-sample relevance values plotted here are used solely as an ex-post descriptive visualization; they never enter the optimization, model selection, hyperparameter choice, or any reported statistical test, all of which use the train-only prior.

3.5. Threshold sensitivity

The 0.5 preferred-bit threshold that separates preferred from disfavored bits (Methods) is a convenience, not a theoretically grounded neutral point. This sweep ($\tau \in \{0.4, 0.5, 0.6\}$, 30 independent runs per cell, same protocol and datasets as the main ablation) tests whether that choice is at least empirically reasonable. No single τ dominates uniformly: $\tau = 0.5$, the value deployed throughout this paper, attains the highest accuracy on two of five datasets; mean selected-feature count falls monotonically as τ increases on every dataset, at a modest, dataset-dependent, non-uniform accuracy cost.

3.6. Comparison with classical selectors

RQ4 asks how robust the mechanism is across classifiers, relevance priors, and dataset dimensionalities; this subsection and Table 7 answer it directly. Table 6 compares RG-SCSO across the full 18-dataset benchmark against five classical filter, embedded, and wrapper selectors – mutual-information thresholding, mRMR, ReliefF, LASSO, and sequential forward selection – and a Concrete-Autoencoder deep feature selector [28], under the identical stratified 5-fold KNN evaluator. RG-SCSO attains the best accuracy of all methods on 12 of 18 datasets. Across the benchmark it is significantly more accurate (Holm-corrected Wilcoxon signed-rank over the 18 datasets) than mutual-information thresholding, sequential forward selection, mRMR, and LASSO (all Holm $p < 0.05$); the one classical selector it does not significantly beat is ReliefF, which it still leads on 13 of 18 datasets ($p = 0.2$). The Concrete-Autoencoder baseline, matched to RG-SCSO’s per-dataset subset size, is the weakest selector on the small-sample datasets tested, consistent with deep reconstruction objectives needing far more than the 50–455 training instances these benchmarks provide. The exceptions to RG-SCSO’s lead are informative and concentrated: on the two gene-expression ($p \gg n$) datasets a classical LASSO baseline attains higher accuracy with far fewer features on both Leukemia (99.8% at 23 features vs. RG-SCSO’s 98.6% at 940) and ColonCancer (95.7% at 64 vs. 88.1% at 563), with mRMR competitive on both; and on four combinatorial or categorical datasets (TicTacToe, M-of-n, KrVsKpEW, WaveformEW) a classical method edges it by at most 1.7 accuracy points. RG-SCSO’s advantage is thus broad but not universal: clearest on datasets without extreme $p \gg n$ structure, where

Table 5: Threshold sensitivity ($\tau \in \{0.4, 0.5, 0.6\}$ replacing the fixed 0.5 preferred-bit threshold in Eq. (2); 30 independent runs per cell, same protocol and datasets as the main ablation). Highest accuracy per dataset in bold. No single τ dominates uniformly: $\tau = 0.5$, the value deployed throughout this paper, attains the highest accuracy on two of five datasets; mean selected-feature count falls monotonically as τ increases on every dataset, at a modest, dataset-dependent, non-uniform accuracy cost. Φ is the Nogueira stability index (Supplementary Information gives the full feature-selection stability comparison across algorithms).

Dataset	τ	Mean Acc.	Mean #Feat.	Φ
ColonCancer	0.4	0.8704	702.5	0.009
	0.5	0.8809	563.1	0.014
	0.6	0.8802	544.5	0.012
Leukemia	0.4	0.9860	1205.4	0.011
	0.5	0.9860	939.7	0.015
	0.6	0.9864	927.6	0.012
Sonar	0.4	0.8920	20.4	0.082
	0.5	0.8940	19.1	0.088
	0.6	0.8916	17.1	0.093
WDBC	0.4	0.9795	13.2	0.193
	0.5	0.9794	11.8	0.190
	0.6	0.9791	11.1	0.163
Zoo	0.4	0.9811	7.5	0.510
	0.5	0.9808	7.3	0.541
	0.6	0.9794	6.3	0.661

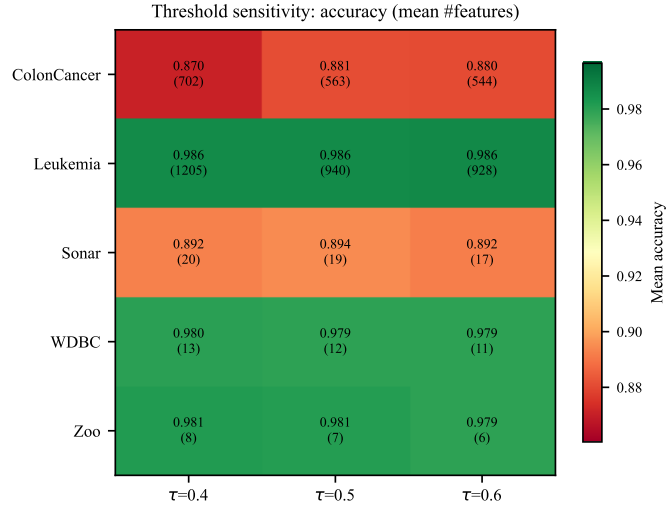


Figure 6: Mean held-out accuracy (color) and mean number of selected features (in parentheses) across the three preferred-bit thresholds tested ($\tau \in \{0.4, 0.5, 0.6\}$), one row per dataset. Accuracy is essentially flat across thresholds on every dataset (largest swing 0.011, ColonCancer $\tau=0.4$ vs. $\tau=0.5$), while the selected-feature count falls monotonically as τ increases on all five datasets, confirming $\tau = 0.5$ is not a fragile choice.

on ultra-high-dimensional gene-expression data a computationally far cheaper embedded method such as LASSO is a strong, arguably preferable, alternative. A remaining question is whether this parsimony mechanism is an artifact of the KNN wrapper used throughout; Table 7 answers it directly, extending the same relevance-guided search to SVM and Random Forest wrappers. The advantage over a no-prior baseline is not a KNN artifact, though it is a more consistent parsimony gain than an accuracy one under Random Forest specifically, and the ReliefF-prior degradation already established above reproduces under every wrapper tested.

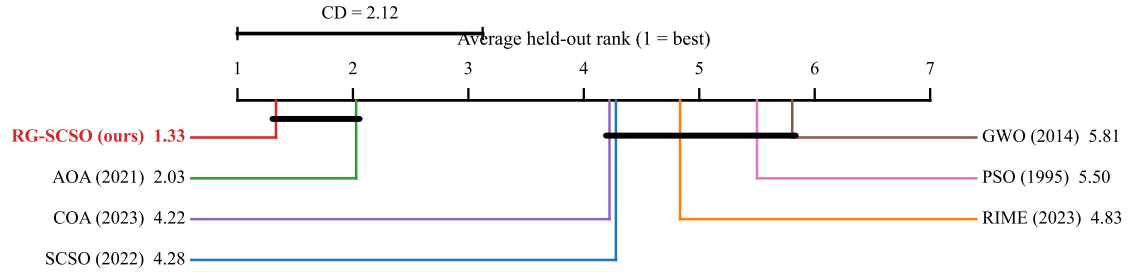


Figure 7: Critical-difference (Nemenyi) diagram at $\alpha = 0.05$ over the held-out ranking (Table 2); algorithms not joined by a bar differ significantly in mean held-out rank. RG-SCSO attains the best mean rank (1.33) and AOA the second (2.03); the remaining five algorithms cluster between 4.31 and 5.75. The critical difference is $CD = 2.12$ (7 algorithms, 18 datasets), so RG-SCSO and AOA are statistically indistinguishable from each other yet both separate from the rest. The corresponding in-sample diagram appears in Supplementary Fig. S6.

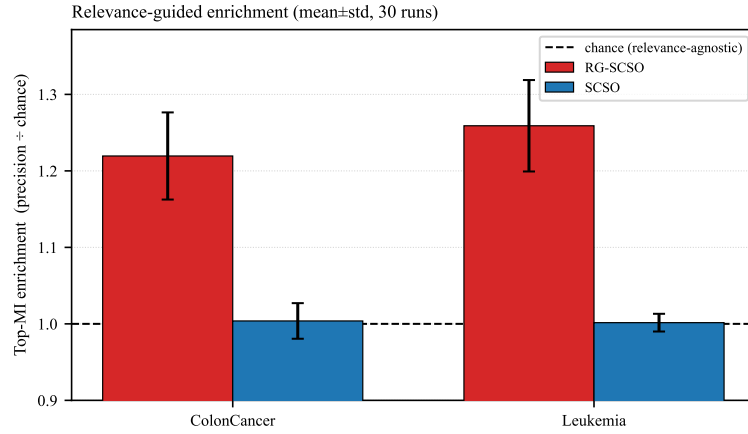


Figure 8: Mechanism evidence: size-fair top-MI enrichment on the two gene-expression sets (selection precision divided by the chance level $|S|/N$; mean over 30 runs, error bars = std). RG-SCSO enriches its subset 1.22–1.26× above chance (ColonCancer, Leukemia respectively), whereas the relevance-agnostic SCSO sits at essentially chance (lift 1.00 on both), evidence consistent with relevance guidance, rather than the base search, contributing to the smaller and more accurate subsets. Feature-selection stability is reported separately, with its size correction, in Fig. 9.

Table 6: Full 18-dataset comparison of RG-SCSO against five classical filter/embedded/wrapper selectors and a Concrete-Autoencoder deep feature selector (CAE), replacing the earlier five-dataset pilot; accuracy with mean selected-feature count in parentheses, 30 runs each, identical evaluator (stratified 5-fold KNN) throughout, bold = best accuracy per dataset. RG-SCSO attains the best accuracy of all methods on 12 of 18 datasets. Across the benchmark it is significantly more accurate (Holm-corrected Wilcoxon signed-rank over the 18 datasets) than ML-thr ($p < 0.001$), SFS ($p = 0.002$), mRMR ($p = 0.017$), LASSO ($p = 0.047$); the one classical method it does not significantly beat is ReliefF (better on 13/18, $p = 0.20$). The CAE baseline, matched to RG-SCSO’s per-dataset subset size, is the weakest selector on the small-sample datasets tested, consistent with deep reconstruction objectives needing more data than these 50–455-instance benchmarks provide. LASSO and mRMR remain the strongest competitors on the two gene-expression ($p \gg n$) sets.

Dataset	RG-SCSO	LASSO	mRMR	ReliefF	SFS	ML-thr	CAE
Diabetes	0.760 (5)	0.735 (8)	0.739 (7)	0.745 (5)	0.741 (7)	0.695 (1)	0.697 (5)
BreastEW	0.974 (4)	0.968 (9)	0.971 (7)	0.972 (6)	0.970 (8)	0.965 (6)	0.946 (4)
TicTacToe	0.820 (7)	0.847 (9)	0.847 (9)	0.847 (9)	0.788 (8)	0.653 (1)	0.738 (6)
HeartDisease	0.849 (5)	0.821 (12)	0.825 (11)	0.829 (10)	0.808 (12)	0.740 (1)	0.753 (5)
M-of-n	0.994 (6)	1.000 (6)	0.965 (8)	1.000 (6)	0.868 (12)	0.664 (1)	0.712 (6)
Vote	0.974 (3)	0.970 (1)	0.970 (1)	0.970 (1)	0.924 (15)	0.968 (1)	0.841 (3)
Zoo	0.981 (7)	0.888 (11)	0.961 (11)	0.965 (11)	0.941 (15)	0.753 (2)	0.871 (6)
Lymphography	0.865 (8)	0.823 (17)	0.836 (13)	0.862 (12)	0.815 (17)	0.695 (1)	0.769 (7)
GermanCredit	0.761 (8)	0.741 (20)	0.744 (15)	0.747 (13)	0.744 (19)	0.673 (1)	0.691 (7)
Parkinsons	0.962 (8)	0.912 (12)	0.924 (9)	0.940 (13)	0.904 (21)	0.867 (1)	0.884 (7)
SpectEW	0.859 (9)	0.810 (14)	0.819 (10)	0.836 (9)	0.800 (21)	0.699 (1)	0.780 (8)
WDBC	0.979 (12)	0.962 (20)	0.972 (17)	0.968 (13)	0.967 (25)	0.943 (10)	0.952 (10)
IonosphereEW	0.914 (8)	0.844 (31)	0.905 (3)	0.899 (4)	0.860 (25)	0.822 (1)	0.862 (7)
KrVsKpEW	0.943 (15)	0.936 (32)	0.923 (26)	0.954 (15)	0.949 (25)	0.522 (1)	0.614 (12)
WaveformEW	0.820 (16)	0.812 (26)	0.786 (23)	0.837 (15)	0.812 (25)	0.503 (1)	0.786 (11)
Sonar	0.894 (19)	0.808 (59)	0.852 (18)	0.820 (8)	0.822 (25)	0.686 (1)	0.787 (16)
ColonCancer	0.881 (563)	0.957 (64)	0.907 (6)	0.907 (11)	0.902 (25)	0.836 (3)	0.794 (324)
Leukemia	0.986 (940)	0.998 (23)	0.989 (6)	0.978 (10)	0.981 (25)	0.963 (30)	0.892 (419)

Table 7: Robustness across classifier wrappers and relevance priors, averaged over five representative datasets (Zoo, Sonar, WDBC, ColonCancer, Leukemia; KNN/SVM use the main study’s 30 independent runs per cell; Random Forest uses a reduced 10, given its far higher per-evaluation cost, disclosed rather than matched artificially). Per-dataset detail for all three wrappers is given in Supplementary Information. Fewest features per wrapper in bold.

Wrapper	Method	Mean Acc.	Mean #Feat.
KNN (30 runs)	RG-SCSO (MI)	0.9442	308.2
	RG-SCSO (ReliefF)	0.9380	565.8
	bSCSO (no prior)	0.9350	545.1
SVM (30 runs)	RG-SCSO (MI)	0.9489	299.1
	RG-SCSO (ReliefF)	0.9476	555.2
	bSCSO (no prior)	0.9468	537.5
RF (10 runs)	RG-SCSO (MI)	0.9521	313.1
	RG-SCSO (ReliefF)	0.9514	571.6
	bSCSO (no prior)	0.9505	546.2

4. Discussion

These results trace washout, a concrete failure mode of transfer-function-based binary feature selection, to its source, and cure it by moving the relevance signal directly inside the binarization operator rather than upstream of it in the objective or the initialization. The formal result motivating this design (Methods) is a diagnostic bound: it explains why continuous-space enhancements fail at the binarization boundary, but it is not a convergence guarantee for RG-SCSO itself. The underlying recipe, a filter prior coupled to a wrapper search with memetic refinement, is a known combination in the feature-selection literature. What is new is the injection point, together with a stringent evaluation protocol: budget-matching, a leak-free hold-out, cross-classifier and cross-prior robustness, and an explicit exploration-safety diagnostic. We report parsimony, not raw accuracy, as the transferable outcome of this choice.

Several boundaries delimit what these results establish. RG-SCSO inherits SCSO’s continuous search dynamics unchanged, and the RMS rule carries a risk of its own: bias the flip probability too strongly and confidently classified bits can freeze in place, draining population diversity. We measured this directly rather than assuming it away: at an aggressive stress-test bias the risk is real and grows with dimensionality, but the conservative $\gamma = 0.5$ this paper deploys keeps the frozen-bit fraction below 1.1% throughout the run on every dataset tested. We also tested whether UMR’s benefit specifically requires targeting relevance-uncertain features, or only the extra evaluation budget it spends, by replacing its targeted K -feature selection with K uniformly random features at matched NFE (Supplementary Information). This untargeted control significantly outperforms targeted UMR on both accuracy and feature count on the two gene-expression datasets, where UMR’s contribution is largest (ColonCancer: 90.6% vs. 88.1% accuracy, 214 vs. 563 features; Leukemia: 423 vs. 940 features at statistically indistinguishable accuracy), and ties it on the other three. UMR’s value over no memetic step at all remains real (Table 3); on the datasets where that value is largest, however, it does not depend on targeting relevance-uncertain features specifically, a more modest claim than “uncertainty-targeted” on its own implies. The relevance field is built from a single filter statistic. Swapping in a ReliefF prior exposes a genuine limitation rather than confirming the mechanism: the parsimony advantage disappears entirely (Supplementary Information). RG-SCSO’s compactness therefore depends on a prior that drives uninformative features below the neutral point, not on the mere presence of a relevance signal. This dependence, together with the $O(dn \log n)$ one-time cost of the mutual-information prior amortized against the wrapper’s $N + K$ per-iteration evaluations, is why we scope the scalability claim to the wrapper cost dominating overall runtime as feature dimensionality grows (full per-dataset wall-clock cost is in Supplementary Information) rather than to any runtime advantage over same-budget baselines, which we do not claim.

The main objective is also a KNN wrapper. Under SVM, tested directly on 16 of the 18 datasets, and under Random Forest on the same five-dataset subset as Table 7, the parsimony advantage is not a KNN artifact. Under Random Forest specifically it is a more consistent gain in subset size than in accuracy, and the ReliefF-prior degradation established above reproduces under every wrapper tested. The Random Forest check is reported at 10 independent runs per cell rather than 30 – a predefined, compute-bound scope decision, as an RF-wrapper run refits a 100-tree ensemble at every fitness evaluation and costs roughly 22× the wall-clock of the KNN wrapper (Supplementary Information) – so it is read here as a directional robustness check rather than a fully powered test. The selected subset is smaller and more consistent in size than competing algorithms’, but not necessarily more consistent in identity. Measured by the size- and chance-corrected Nogueira Φ (Fig. 9a), RG-SCSO exhibited more size-corrected feature-selection stability than the baseline SCSO across the evaluated datasets, clearest on the three lower-dimensional sets, yet on both gene-expression datasets RG-SCSO’s own Φ is itself close to the level expected by chance, so a much smaller subset there is not a materially more repeatable one. The size correction matters here: an uncorrected pairwise Jaccard would rank AOA as the most “stable” method only because it selects almost every feature every run (Fig. 9b), overlap that is an artifact of subset size rather than genuine repeatability; the raw-Jaccard values are reported as a metric-diagnostic in Supplementary Information to make this behaviour explicit rather than to headline a favourable number. The full per-dataset breakdown for both checks is in Supplementary Information.

This wrapper search is itself compute-intensive; absolute wall-clock cost per run, for every algorithm

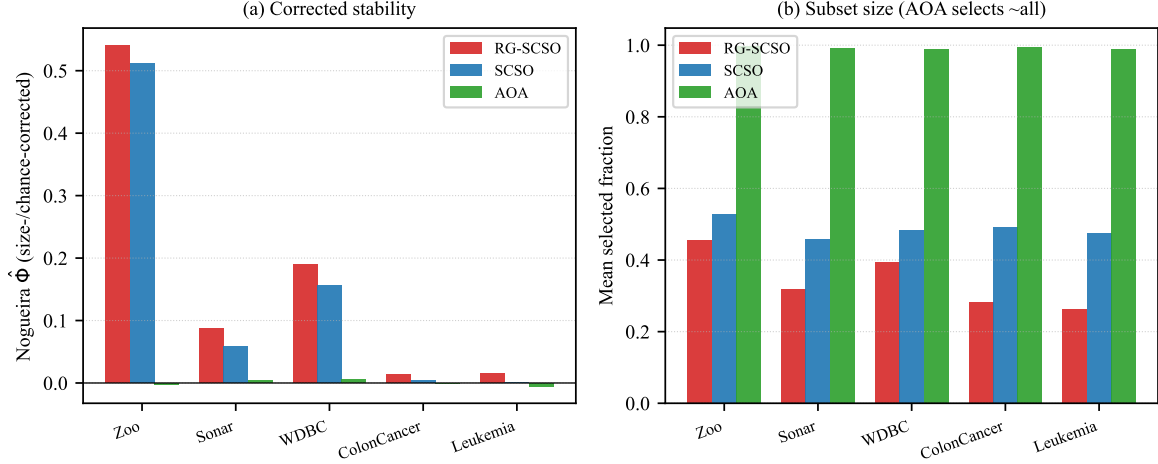


Figure 9: Feature-subset stability across the five representative datasets, 30 runs each. (a) Size- and chance-corrected Nogueira $\hat{\Phi}$: RG-SCSO exhibited more size-corrected run-to-run stability than the baseline SCSO on every evaluated dataset, while AOA sits at $\hat{\Phi} \approx 0$ (no better than chance). (b) Mean selected fraction, explaining why an uncorrected metric misleads: AOA selects ≈ 98 – 99% of all features, so its subsets trivially overlap, whereas RG-SCSO selects the fewest. Raw pairwise Jaccard (confounded by this size imbalance) is provided only as a Supplementary diagnostic.

tested, is reported in full in Supplementary Information rather than only argued qualitatively. A genuine outer-fold nested cross-validation pilot on three datasets, also in Supplementary Information, is directionally consistent with the single-split held-out estimate above, though underpowered at five runs per cell to confirm significance independently.

External validity has its own limits. Generalization here is assessed in the sense appropriate to a supervised feature-selection benchmark – held-out instances, cross-classifier, cross-prior, and cross-dimensionality evaluation; unseen-class generalization is not an applicable estimand for this task and is therefore not tested. The benchmark spans 8 to 3571 features across biomedical, gene-expression, and categorical domains drawn from a single curated family of UCI and standard microarray sets. Its top end (3571 features, Leukemia) is still an order of magnitude below the 10^4 – 10^5 -feature omics and high-dimensional text regimes where a binary-native, relevance-guided operator should matter most, so behaviour at that scale is extrapolated, not measured; a genuine $>10^4$ -feature evaluation is deferred to future work because a full seven-optimizer, 30-run wrapper study at that dimensionality is compute-bound by the $O(N \max_nfe)$ KNN fitness evaluations rather than by the relevance mechanism itself, and lies outside this study’s fixed budget. The relevance field itself is not perfectly stable on the smallest, highest-dimensional datasets: a bootstrap resampling analysis, reported in Supplementary Information, shows markedly lower rank stability of the mutual-information field on the two gene-expression sets than on the three lower-dimensional benchmarks, a risk the classical-baseline comparison above makes concrete rather than theoretical.

Finally, the accuracy claim is scoped, not universal: against binary particle-swarm and grey-wolf optimizers carrying published adaptive transfers, and against same-family binary SCSO selectors, RG-SCSO does not lead on accuracy, and on the two gene-expression datasets specifically a classical LASSO baseline outperforms RG-SCSO outright, so we do not claim a practical advantage for RG-SCSO on ultra-high-dimensional, small-sample data; we scope its transferable benefit to parsimony on datasets without that extreme structure. A further boundary is that the baselines are evaluated at their published/default control parameters under a shared protocol rather than individually re-tuned per dataset; this is a faithful-reproduction comparison, and a per-dataset tuning study of every competitor is left to future work. A final, deliberate scope boundary concerns the host optimizer: the robustness evaluated here is across classifiers, relevance priors, and dataset dimensionalities (RQ4), all within SCSO. Whether the same per-feature, relevance-modulated binarization operator transfers its benefit to other swarm optimizers was not evaluated in the present study and therefore remains an open direction for future work (Future work, item iii); accordingly we make no

cross-optimizer generalization claim. These boundaries are gathered together, with the future work they motivate, in Conclusion below.

5. Conclusion

The results support RG-SCSO as a relevance-guided binary feature-selection method whose primary advantage is improved subset parsimony with competitive predictive performance under the tested conditions: across 18 benchmark datasets it selects the second-smallest feature subsets of any method evaluated, trailing only COA, while attaining the best mean held-out accuracy among all seven algorithms compared, under a fixed-budget, leak-free evaluation protocol (RQ1, RQ2).

This advantage is not universal, and the boundaries established in Discussion above are real constraints on the claim, not caveats to be read past. Against optimizers carrying published adaptive continuous-space transfers, and against same-family binary SCSO selectors, RG-SCSO does not lead on accuracy. On the two extreme $p \gg n$ gene-expression datasets tested, a classical LASSO baseline outperforms it outright on both accuracy and feature count. Its dependence on the underlying relevance prior is genuine: replacing mutual information with a ReliefF prior removes the parsimony advantage entirely. The 0.5 preferred-bit threshold, while empirically reasonable across the range tested, is not a uniformly optimal choice. And the benchmark itself, though spanning 8–3571 features across biomedical, gene-expression, and categorical domains, is drawn from a single curated family of datasets, so behavior on ultra-high-dimensional omics data of 10^4 – 10^5 features is extrapolated rather than measured.

In practical terms, RG-SCSO is designed to drop into the binarization step of an SCSO-based wrapper, adding only a small, fixed set of hyperparameters (the modulation strength γ , the preferred-bit threshold τ , and the memetic budget K) that this paper shows are not fragile choices; its main practical payoff is a smaller feature subset that is cheaper to store and can lower downstream feature-processing cost wherever feature acquisition and preprocessing scale with dimensionality, at accuracy that matches or exceeds the swarm-based selectors it was compared against.

Future work includes: (i) adaptive, data-driven selection of the preferred-bit threshold, for example via cross-validation or an entropy-based criterion, rather than a fixed value; (ii) an explicit multi-objective formulation that traces an accuracy-parsimony Pareto front rather than a single weighted objective; (iii) extending the relevance-modulated binarization mechanism to other swarm optimizers, such as WOA, HHO, or SMA, to test whether the injection-point argument generalizes beyond SCSO; (iv) evaluation at larger scale, on datasets exceeding 10^4 features, where a relevance-guided, binary-native operator should be especially valuable; and (v) alternative relevance priors beyond mutual information and ReliefF, such as SHAP-based, mRMR, or learned priors, to test whether the mechanism’s benefit is specific to the prior used here or transfers to others.

6. Methods

6.1. Problem formulation

Let a candidate subset be a binary mask $b \in \{0, 1\}^d$. The objective is

$$f(b) = 0.99(1 - \text{Acc}(b)) + 0.01 \frac{|b|}{d}, \quad (1)$$

where $\text{Acc}(b)$ is the stratified 5-fold KNN accuracy ($k = 5$) using only the selected features. Every baseline in this study optimizes this same objective with the same 0.01 cardinality weight. RG-SCSO is granted no structural advantage on subset size from the fitness function itself, so any parsimony gap reported below reflects the search mechanism, not a differently weighted objective. SCSO governs its search with the sensitivity range $R(t) = S_M - S_M t/T$ ($S_M = 2$), which RG-SCSO retains while replacing the binarization and adding a relevance field.

Algorithm 1 RG-SCSO for binary feature selection

Require: training partition (X, y) (never the held-out fold; see Methods below for the fold-honest protocol); population size N ; iterations $T_{\max}$; budget max_nfe ; memetic size K ; bias strength γ

Ensure: best feature mask b^*

- 1: $\rho_j \leftarrow$ normalized mutual information $I(X_j; y)$, computed on (X, y) only, $\forall j$ ▷ static relevance field
- 2: $\forall j$: preferred bit $\hat{b}_j \leftarrow \mathbf{1}[\rho_j > 0.5]$; strength $s_j \leftarrow 2|\rho_j - 0.5|$
- 3: initialize positions $x_i \sim \mathcal{U}(-1, 1)^d$, $i = 1, \dots, N$; form the initial masks by sign only ($b_i = 1$ if $x_i > 0$, else 0); evaluate; set b^* ▷ the relevance-modulated flip rule (lines 8–11) is not applied at initialization
- 4: while $\text{nfe} < \text{max_nfe}$ do
- 5: $R \leftarrow S_M (1 - t/T_{\max})$ ▷ sensitivity range contracts
- 6: for each agent $i = 1, \dots, N$ do
- 7: update x_i by the SCSO position rule using range R
- 8: for each feature j do
- 9: $p_{\text{base}} \leftarrow |\tanh(x_{ij})|$ ▷ base transfer $T(x_{ij})$
- 10: $p_j \leftarrow p_{\text{base}} (1 + \gamma s_j)$ if the flip moves bit j toward $\hat{b}_j$, else $p_{\text{base}} (1 - \gamma s_j)$
- 11: flip bit j with probability $\text{clip}(p_j, 0, 1)$
- 12: end for
- 13: evaluate mask; update b^* if improved
- 14: end for
- 15: $U \leftarrow K$ features whose ρ_j is closest to 0.5 ▷ UMR on uncertain bits
- 16: for each $j \in U$ do
- 17: flip bit j of b^* ; keep the flip only if fitness improves
- 18: end for
- 19: end while
- 20: return b^*

6.2. Theoretical motivation: transfer-function washout

Consider any binary-native optimizer that keeps a real-valued position and binarizes coordinate j through a transfer $T : \mathbb{R} \rightarrow [0, 1]$. A continuous-space enhancement can influence the retained subset only by perturbing a coordinate, $x_j \mapsto x_j + \delta_j$; its entire effect on the discrete decision is the induced change in probability $\Delta_j = T(x_j + \delta_j) - T(x_j)$.

Lemma 1 (Leverage bound). Let $T : \mathbb{R} \rightarrow [0, 1]$ be Lipschitz and piecewise continuously differentiable. If a coordinate-space enhancement perturbs x_j by δ_j , $|\Delta_j| \leq \|T'\|_{\infty} |\delta_j|$, where $\|T'\|_{\infty}$ is the largest slope T attains between x_j and $x_j + \delta_j$. For the two standard transfers $\|\sigma'\|_{\infty} = \frac{1}{4}$ and $\|\tanh'\|_{\infty} = 1$, and each slope decays away from the origin; in a flat region, where $\|T'\|_{\infty} \leq \varepsilon$, the leverage collapses to $|\Delta_j| \leq \varepsilon |\delta_j|$ (proof in Supplementary Information).

Remark (why RG-SCSO is exempt). RG-SCSO breaks the premise of the lemma: instead of routing information through the coordinate, it modulates the flip probability directly at the transfer output, $\Delta_j = \pm \gamma s_j V(x_j)$, independent of T' . The base transfer T still appears inside RG-SCSO’s own rule as one multiplicative factor (Eq. 2 below), but Proposition 2, which follows next, characterizes only the base operator’s transition probability $T(x_j^{(t)})$ itself, not RG-SCSO’s realized flip probability p_j , which this Remark has just shown is exempt from Lemma 1’s bound by construction.

Proposition 2 (Transition probability under the base V-shaped transfer). This proposition characterizes only the base, unmodulated V-shaped transfer introduced above (the NoRMS reference operator used in the ablation); RG-SCSO’s own operator is exempt from it by construction, per the Remark above. Under this unmodulated V-shaped binarization rule, bit j flips at iteration t with probability $T(x_j^{(t)})$, so the base operator’s one-step bit-transition probability coincides with T , $P_{\text{base}}(b_j^{(t+1)} \neq b_j^{(t)} \mid x_j^{(t)}) = T(x_j^{(t)})$. Lemma 1’s bound on Δ_j is therefore, without further assumption, already a bound on the one-step change in the base operator’s transition probability, and this bound composes linearly over N repeated samples of the same bit in

the flat, saturated regime (proof in Supplementary Information). For RG-SCSO itself, this base probability $T(x_j^{(t)})$ is subsequently modulated by the relevance-modulated rule p_j of Eq. 2 below, so RG-SCSO’s actual one-step transition probability is p_j , not $T(x_j^{(t)})$ alone. This is a local, one-step-composable sensitivity bound, not a convergence guarantee or a proof that continuous-space enhancements are ineffective; the full scope discussion is in Supplementary Information.

6.3. The RG-SCSO mechanism

Given the updated position x_j , the base flip probability uses a V-shaped transfer $V(x_j) = |\tanh(x_j)|$. With relevance $\rho_j \in [0, 1]$, preferred bit $b_j^* = \mathbf{1}[\rho_j > 0.5]$ and strength $s_j = 2|\rho_j - 0.5| \in [0, 1]$, the flip probability is biased toward b_j^* ,

$$p_j = \text{clip}(|\tanh(x_j)| (1 + \gamma \sigma_j s_j), 0, 1), \quad (2)$$

where $\sigma_j = +1$ if the flip moves bit j toward b_j^* and $\sigma_j = -1$ otherwise. Setting $\gamma = 0$ recovers a plain V-shaped operator (the NoRMS ablation). The 0.5 threshold that separates preferred from disfavored bits is a convenience, not a theoretically grounded neutral point: with $\rho_j = I(X_j; y)/H(y)$, the value at which a feature stops being informative depends on the number of classes, the label entropy, the sample size, and the specific MI estimator used, none of which are normalized away by the $H(y)$ division alone. The Discussion shows this dependence is not merely theoretical: the method’s compactness is tied to how the chosen relevance statistic happens to place features around 0.5, not to the threshold itself. A threshold sweep ($\tau \in \{0.4, 0.5, 0.6\}$, Results Section 3.4) supports this choice as reasonable rather than arbitrary: $\tau = 0.5$ attains the highest accuracy on two of five datasets tested, and every value trades a modest, dataset-dependent accuracy shift for a monotonic reduction in feature count as τ increases, with no single value dominating uniformly.

The field ρ that drives this mechanism is, in the final method, the static prior $\rho_{\text{static}} = \text{clip}(\text{MI}(f_j; y)/H(y), 0, 1)$, a normalized mutual-information filter score [2, 29] computed once from the data; an online extension adding an EMA credit-assignment term from accepted fitness improvements was examined and dropped, as the ablation shows no accuracy gain on any dataset. A second, smaller component, uncertainty-targeted memetic refinement (UMR), spends a fixed local-search budget where this prior is least decisive: each iteration, the K features whose relevance is closest to 0.5 are greedily flipped on the incumbent best mask [15] and the flip is kept only if fitness improves.

6.4. Algorithm and computational cost

Every fitness evaluation, whether from population moves, memetic probes, or initialization, is counted against a single budget $\text{max_nfe} = 15000$, so UMR grants no extra evaluations (Algorithm 1). This cap is enforced identically for every algorithm: RG-SCSO and the base SCSO stop as soon as the counter reaches 15,000, and each population-based baseline is run under a hard function-evaluation termination at the same 15,000. The only residual difference is CoatiOA, which evaluates in sub-population batches and therefore halts at the first batch boundary not below the cap (15,030, a 0.2% excess that favours the baseline, not RG-SCSO). Budgets are thus matched to within 0.2% across all seven methods, removing any budget advantage for the proposed method.

The per-iteration cost is dominated by the $N + K$ wrapper evaluations, each a KNN fit under fixed folds; the mutual-information prior adds a one-time $O(dn \log n)$ preprocessing cost (the nearest-neighbor MI estimator [29]), amortized against the wrapper evaluations. RG-SCSO is thus asymptotically no more expensive than base SCSO apart from the K extra probes. In wrapper feature selection generally, classifier-based fitness evaluation, not the search mechanism itself, dominates overall computational cost as feature dimensionality grows; per-dataset wall-clock measurements across the full 8–3571-feature range tested (Supplementary Information) are consistent with this expectation, and we make no runtime-speedup claim for RG-SCSO over same-budget baselines.

6.5. Datasets, baselines, and protocol

The benchmark spans 18 preprocessed datasets of 8 to 3571 features across biomedical, gene-expression, and categorical domains, full characteristics in Supplementary Table S1.

Leakage control (data flow).. Every step that could learn from the labels or the held-out partition is confined to the training fold. The flow is:

raw data $\rightarrow$ label-independent cleaning/encoding $\rightarrow$ outer 80/20 split $\rightarrow$ [on the 80% train only:] fit standardizer $\cdot$
 compute MI relevance prior ρ $\cdot$ 5-fold CV fitness $\cdot$ feature-subset search $\rightarrow$ freeze selected mask $\rightarrow$ [on the 20%
 hold-out:] transform with the train-fitted standardizer only $\rightarrow$ single KNN evaluation

No standardizer, relevance prior, cross-validation fold, or hyperparameter is ever fit on, or given access to, the held-out partition; the hold-out is touched exactly once, for final evaluation.

We benchmark against six baselines: SCSO (base) [10], AOA [6], CoatiOA [7], GWO [30], PSO [4], and RIME [8]. The protocol is pre-specified and locked prior to the full run. All algorithms share: population 30, 500 iterations, 30 independent runs, seed = 42 + run_id (paired across algorithms), KNN ($k = 5$) with stratified 5-fold cross-validation, search space $[-1, 1]^d$, and the fitness of (1). This is a faithful published-default comparison, not an equally-tuned one: every baseline runs at its own published/library-default control parameters under this identical shared protocol, and we do not re-tune any method per dataset. RG-SCSO’s $\gamma = 0.5$, $\tau = 0.5$, and $K = 8$ are likewise fixed before the full run; τ and the cardinality weight λ are additionally characterized by the sensitivity sweeps reported in Supplementary Information, while γ and K were frozen from the pre-specification pilot and not tuned on the test data. A per-method fidelity table (source, version, control parameters, search space, tuning, budget, seed) is provided in the released repository (Baseline_Fidelity_Report); the limitation of comparing against untuned baselines is acknowledged in Discussion.

6.6. Statistics and reproducibility

Statistical units.. We state the inference units explicitly. The experimental unit is one independent run (an outer 80/20 split under a distinct seed). The primary cross-algorithm comparison is at the dataset level: for each dataset and algorithm the 30 runs are aggregated to a mean held-out accuracy, giving 18 paired per-dataset summaries that are the units for the Friedman test and the critical-difference diagram, and for the paired bootstrap confidence intervals of the accuracy difference (RG-SCSO–baseline). The per-dataset Wilcoxon signed-rank test instead treats the 30 paired runs within a single dataset as its unit; these within-dataset results are therefore repeated-run comparisons, not cross-dataset generalization, and are reported as such. Cohen’s d and rank-biserial r are computed on the same paired samples as the test they accompany.

Significance uses the paired Wilcoxon signed-rank test with Holm correction [31], the Holm family formed per dataset, at $\alpha = 0.05$. A tie denotes failure to reject H_0 (no significant accuracy difference) at this threshold; it is not evidence of equivalence. Effect sizes use Cohen’s d and rank-biserial r ; overall comparison uses the Friedman test with a critical-difference diagram [32]. The experimental design was pre-specified and version-controlled before the full run and left unmodified after results were observed; all randomness is seeded deterministically and shared across algorithms.

Author Contributions

Bui Quang Huy: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Writing – original draft, Writing – review & editing, Visualization, Supervision. Duong Minh Son: Formal analysis, Writing – review & editing.

Statements and Declarations

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Conflict of interest

The authors declare no competing interests.

Ethical approval

Not applicable. This study uses only publicly available benchmark datasets (UCI and standard microarray sets) and involves no human participants, human data, or animals.

Use of AI tools

The authors used a generative AI assistant for language editing, code refactoring, and formatting of tables and figures. All experimental design, implementation, results, and scientific claims were produced and verified by the authors, who take full responsibility for the content of this manuscript.

Data Availability Statement

The datasets analysed in this study are publicly available benchmarks (UCI and standard microarray sets); Supplementary Table S1 lists each source. The source code, the locked pre-specification, the per-run seeds, a complete hyperparameter table, a pinned dependency list (requirements.txt), an MIT LICENSE, and the raw per-run results are available for review in an anonymized repository (<https://anonymous.4open.science/r/RG-SCSO>) at the manuscript's release commit, and will be deposited in a public, citable repository with a permanent Zenodo DOI upon acceptance. With the pinned dependencies (requirements.txt, Python 3.10) and the per-run seeds, the tables and figures can be regenerated from the released raw results, and the experiments re-run deterministically.

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